

Integration - Set 2 - Solutions

Specification MM7.1-MM7.6.

Answer key

Question	Answer	Topic
1	D	Integration
2	D	Integration
3	B	Integration
4	F	Integration
5	E	Integration
6	E	Integration
7	A	Integration
8	D	Integration
9	E	Integration
10	D	Integration

Worked solutions

Question 1 INT-18 • Integration • spec MM7.3

Let $I = \int_0^1 (x^2 + 1) dx$ and $J = \int_0^1 (x + 1)^2 dx$. Which of the following is true?

Answer: D

$$I = \frac{1}{3} + 1 = \frac{4}{3}, J = \int_0^1 (x^2 + 2x + 1) dx = \frac{1}{3} + 1 + 1 = \frac{7}{3}, J - I = 1.$$

Question 2 INT-20 • Integration • spec MM7.2

Evaluate $\int_{-1}^2 |x| dx$.

Answer: D

$$\text{Two triangles: } \frac{1}{2} \cdot 1 \cdot 1 + \frac{1}{2} \cdot 2 \cdot 2 = \frac{1}{2} + 2 = \frac{5}{2}.$$

Question 3 X2-17 • Integration • spec MM7.5

The trapezium rule with n strips is used to estimate $\int_0^1 \sqrt{x} dx$. Which of the following is true for every n ?

Answer: B

$y = \sqrt{x}$ is concave (bends downwards), so every chord lies below the curve and the trapezia underestimate the area.

Question 4 INT-01 • Integration • spec MM7.2

Evaluate $\int_1^4 \frac{3x^2 - 5}{\sqrt{x}} dx$.

Answer: F

$$\begin{aligned} \text{Integrand} &= 3x^{3/2} - 5x^{-1/2}; \text{ integral} \\ &= \left[\frac{6}{5}x^{5/2} - 10x^{1/2} \right]_1^4 = \left(\frac{6 \cdot 32}{5} - 20 \right) - \left(\frac{6}{5} - 10 \right) = \frac{192}{5} - 20 - \frac{6}{5} + 10 = \frac{186}{5} - 10 = \frac{136}{5}. \end{aligned}$$

Question 5 INT-05 • Integration • spec MM7.1

Find the area of the finite region enclosed by the curve $y = x^2$ and the line $y = 2x + 3$.

Answer: E

$$\begin{aligned} \text{Intersections: } x^2 - 2x - 3 &= (x - 3)(x + 1) = 0. \text{ Area} \\ &= \int_{-1}^3 (2x + 3 - x^2) dx = \left[x^2 + 3x - \frac{x^3}{3} \right]_{-1}^3 = (9 + 9 - 9) - \left(1 - 3 + \frac{1}{3} \right) = 9 + \frac{5}{3} = \frac{32}{3}. \end{aligned}$$

Question 6 INT-11 • Integration • spec MM7.1

The region bounded by $y = 4 - x^2$, the x -axis and the lines $x = 0$ and $x = 3$ is shaded. Find the total shaded area.

Answer: E

$$\begin{aligned} \text{From 0 to 2 the curve is above the axis: } \int_0^2 (4 - x^2) dx &= 8 - \frac{8}{3} = \frac{16}{3}. \text{ From 2 to 3 it is below:} \\ \left| \int_2^3 (4 - x^2) dx \right| &= \left| (12 - 9) - \frac{16}{3} \right| = \frac{7}{3}. \text{ Total } \frac{23}{3}. \end{aligned}$$

Question 7 INT-14 · Integration · spec MM7.5

The trapezium rule is used to estimate $\int_a^b f(x) dx$. For which of the following functions is the estimate guaranteed to be an underestimate for every choice of $a < b$?

Answer: A

The trapezium rule underestimates when the curve is concave (bends downwards) so that chords lie below the curve: $y = -x^2$ everywhere. x^3 changes concavity at 0.

Question 8 INT-15 · Integration · spec MM7.1

The area between the curve $y = x^2$ and the line $y = k$ (where $k > 0$) is $\frac{32}{3}$. Find k .

Answer: D

$$\text{Area} = \int_{-\sqrt{k}}^{\sqrt{k}} (k - x^2) dx = 2 \left(k^{3/2} - \frac{k^{3/2}}{3} \right) = \frac{4}{3} k^{3/2} = \frac{32}{3} \Rightarrow k^{3/2} = 8 \Rightarrow k = 4.$$

Question 9 INT-19 · Integration · spec MM7.1

The curve $y = x(x-1)(x-2)$ and the x -axis enclose two regions. What is the ratio of the area of the region for $0 \leq x \leq 1$ to the area for $1 \leq x \leq 2$?

Answer: E

The cubic has rotational symmetry about $(1, 0)$ (it is $u^3 - u$ with $u = x - 1$, an odd function), so the two regions are congruent: ratio 1 : 1.

Question 10 X2-03 · Integration · spec MM7.1

f is a function defined for all real x and $\int_0^4 f(x) dx = 0$. Which of the following statements must be true?

- I $f(x) = 0$ for some x with $0 \leq x \leq 4$.
- II The area between the curve and the x -axis from 0 to 4 is 0.
- III $\int_0^2 f(x) dx = -\int_2^4 f(x) dx$

Answer: D

III is just the rule for combining integrals over adjacent ranges. I fails for a function that jumps from -1 to 1 without passing through 0 (nothing says f is continuous). II fails: $f(x) = x - 2$ has integral 0 but area 4.